



Digital Forensics Internship

Windows User Activity Forensics Investigation

Forensic Investigation Report

Week 1 - Task 1

Case ID: PKCERT-DFIR-2026-001

Case Title: Insider Misuse Investigation

Incident:

ABC Technologies suspected that an employee copied confidential company documents to a USB storage device before leaving the organization. A forensic examination was conducted on the employee's Windows 10 workstation to reconstruct user activity during the last 48 hours and determine whether confidential information was accessed, copied, or deleted.

Submitted by: Group No 10

- Muhammad Ubaid
- Imtiaz Ali

Investigation Date: Thursday, July 2026

Operating System Examined: Microsoft Windows 10 (Virtual Machine)

Tools Used:

- Autopsy
- FTK Imager
- WinPrefetchView
- Event Viewer

Confidentiality:**Training Case**

This report has been prepared for educational purposes as part of the PKCERT Digital Forensics Internship. The incident, organization, and user activities described in this report are part of a controlled laboratory exercise and do not represent a real-world investigation.

Acknowledgement:

We would like to express our sincere gratitude to **PKCERT** and our internship supervisor for providing us with the opportunity to perform this digital forensic investigation as part of our training. This project allowed us to gain practical experience in analyzing Windows forensic artifacts, reconstructing user activity, and preparing a professional forensic investigation report.

We also appreciate the guidance and resources provided throughout the internship, which helped us understand the digital forensic investigation process and the importance of preserving and analyzing digital evidence.

Finally, we thank our team members for their cooperation and contribution during the completion of this project.

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1. Case Description

ABC Technologies suspected that an employee attempted to copy confidential company information before leaving the organization. According to the case scenario, the employee denied any unauthorized activity. The company's IT department secured the employee's Windows 10 computer and created a forensic image for examination.

The objective of this investigation is to examine the forensic image, reconstruct the user's activity during the last 48 hours, and identify evidence related to application execution, document access, USB device usage, browser activity, and deleted files.

2. Objectives

The objectives of this forensic investigation are:

- Reconstruct user activity during the last 48 hours.
- Identify user logon and logoff events.
- Determine which applications were executed.
- Identify recently opened files and folders.
- Examine USB device activity.
- Identify browser usage and downloaded files.
- Recover evidence of deleted files.
- Build a chronological timeline of user activity.

3. Evidence Acquisition

3.1 Evidence Acquisition

The Windows 10 virtual machine used in this investigation was acquired using **FTK Imager**. A forensic image of the system drive was created in **Expert Witness Format (E01)**. During the acquisition process, hash values were generated and verified to ensure that the forensic image was an exact copy of the original evidence. Hash verification confirmed that the image remained unchanged throughout the acquisition process, preserving the integrity of the evidence for analysis.

3.2 Evidence Details

Item	Value
Evidence ID	E-001
Evidence Description	Windows 10 Virtual Machine Disk Image
Evidence Format	Expert Witness Format (.E01)
Acquisition Tool	FTK Imager
Operating System	Windows 10
Image Verification	Successful
Acquisition Date	09 July 2026
Examiner	Muhammad Ubaid ,Imtiaz Ali

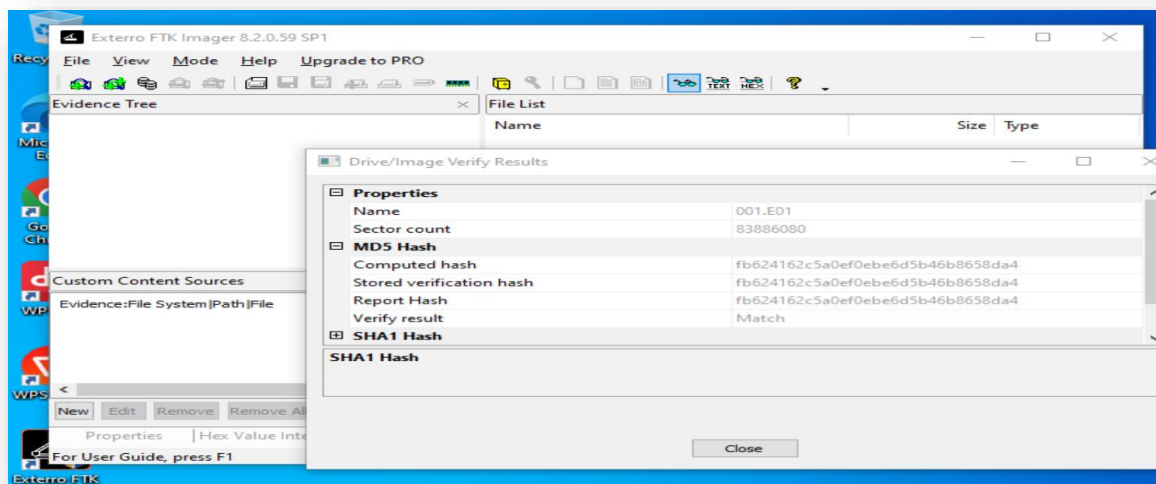
The screenshot shows the 'Create Image' dialog box in FTK Imager. The 'Image Source' tab is selected, and the 'Evidence Item Information' sub-dialog is open. The sub-dialog contains the following fields and values:

- Case Number: PKCERT-DFIR-2026-001
- Evidence Number: 001
- Unique Description: Windows user Activity Investigation
- Examiner: M.Ubaid, Imtiaz Ali
- Notes: Week 1 Task Internship

At the bottom of the sub-dialog are buttons for '< Back', 'Next >', 'Cancel', and 'Help'. Below the sub-dialog, there are 'Start' and 'Cancel' buttons. The background shows a portion of the FTK Imager interface with a table header 'Size | Typ'.

3.3 Hash Verification

Algorithm	Hash Value
MD5	fb624162c5a0ef0ebe6d5b46b8658da4
SHA1 or SHA-256	770db41927a418b59d78aaac6eee4cbe24f6aeb6



3.4 Evidence Integrity

The integrity of the acquired forensic image was verified using cryptographic hash values generated by FTK Imager. The verification process confirmed that the acquired image matched the original source, indicating that no changes occurred during acquisition. This ensures that the evidence examined during the investigation is authentic and reliable.

4. Investigation Methodology

The forensic image was examined using a structured forensic process. The investigation focused on reconstructing the user's activities during the last 48 hours before the system was seized.

The following steps were performed during the examination:

- Verified the integrity of the forensic image using cryptographic hash values.
- Opened the forensic image in Autopsy for analysis.
- Examined Windows user profiles and file system structure.
- Analyzed Windows Event Logs to identify user logon and logoff activity.
- Examined Prefetch files to determine executed applications.
- Reviewed UserAssist, RunMRU, and RecentDocs registry artifacts.
- Investigated USB device connection history.
- Examined browser artifacts for browsing and download activity.
- Reviewed Recycle Bin and deleted files.
- Correlated evidence from multiple artifacts to reconstruct a chronological timeline of user activity.

5. Forensic Analysis

5.1 User Profile Examination

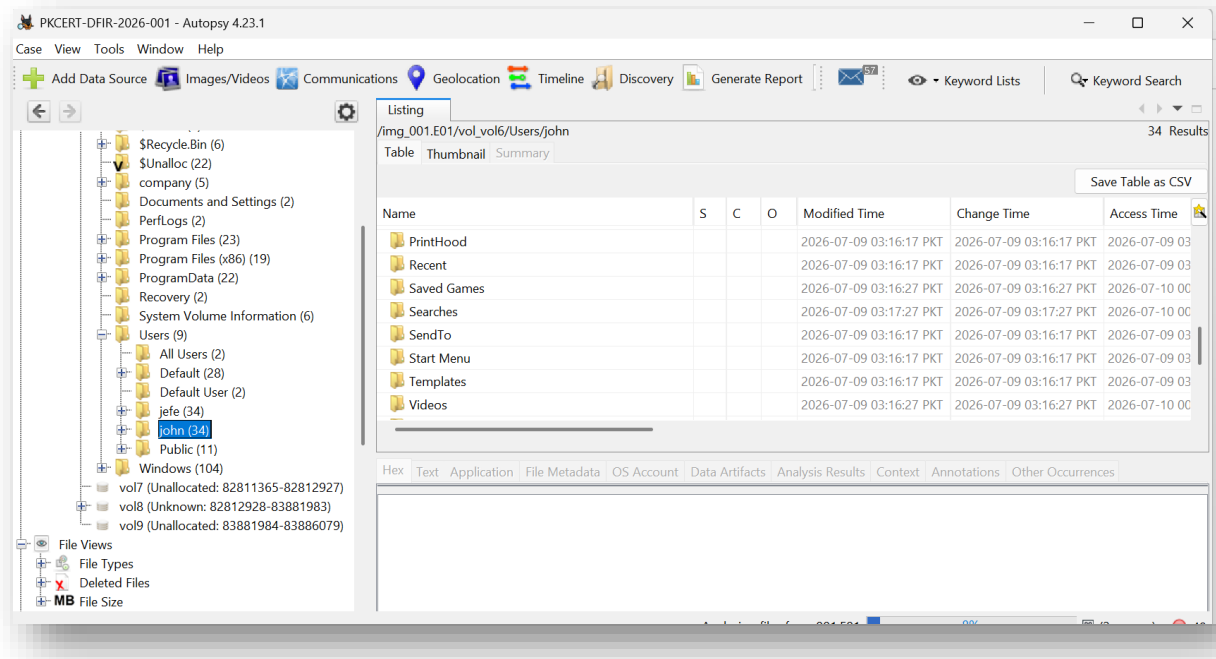
Purpose

To identify the primary user profile and verify the account used during the investigation period.

Location

C:\Users\

Evidence



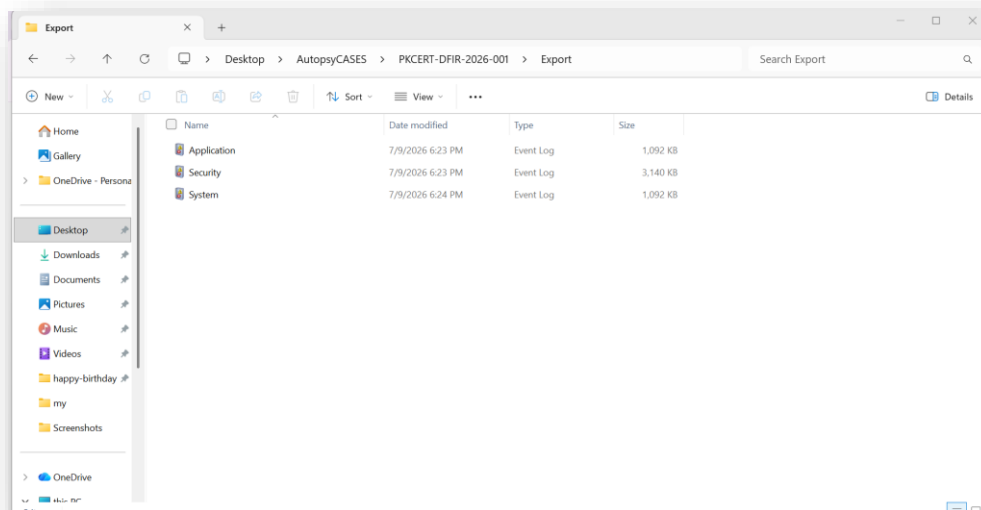
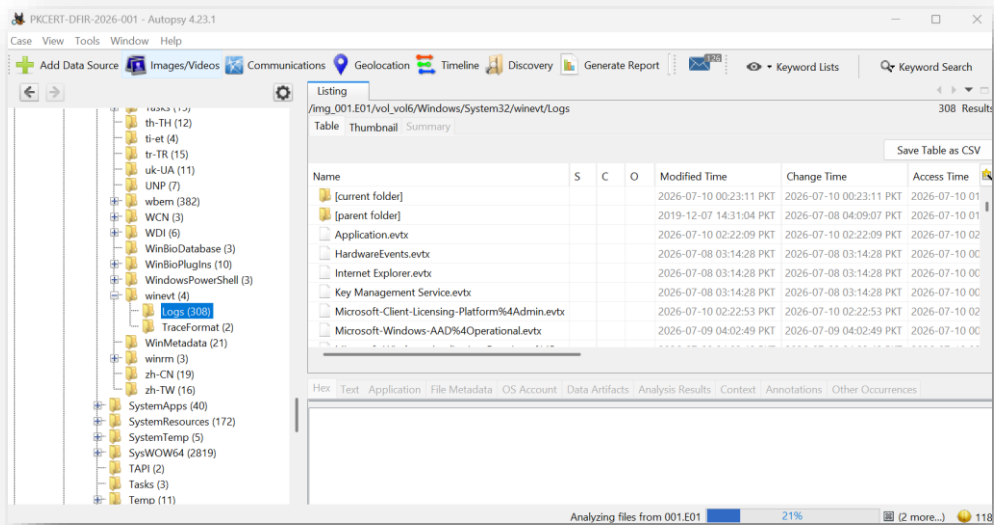
Analysis

The examination identified 2 primary user account used on the system. The profile contains standard user folders such as Desktop, Documents, Downloads, Pictures, and AppData, indicating regular interactive use of the account.

5.2 — Windows Event Log Analysis

Purpose

After identifying the user profile on the system, the next step was to determine what activities occurred during the incident period. Windows Event Logs were examined to establish the timeline of user login, logoff, system startup, shutdown, and device-related events during the last 48 hours. This timeline serves as the foundation for correlating evidence from other forensic artifacts.



Evidence Source

- Artifact: Windows Security Event Log (Security.evtx)
- Tool Used: Microsoft Event Viewer
- Events Examined:
 - Event ID **4624** – Successful Logon
 - Event ID **4647** – User Initiated Logoff
 - Event ID **4634** – Logoff (checked but no corresponding event for this session was identified)

Findings

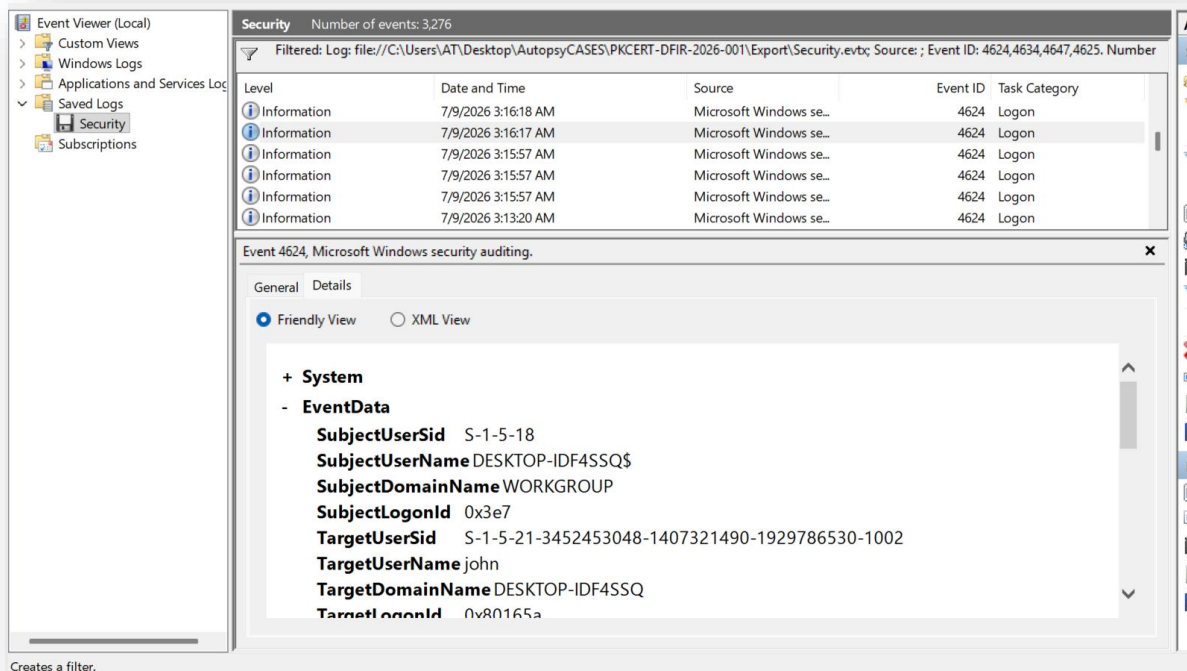
The Security Event Log shows that user **john** successfully logged into the system on **09 July 2026 at 03:16 AM (Local Time)**.

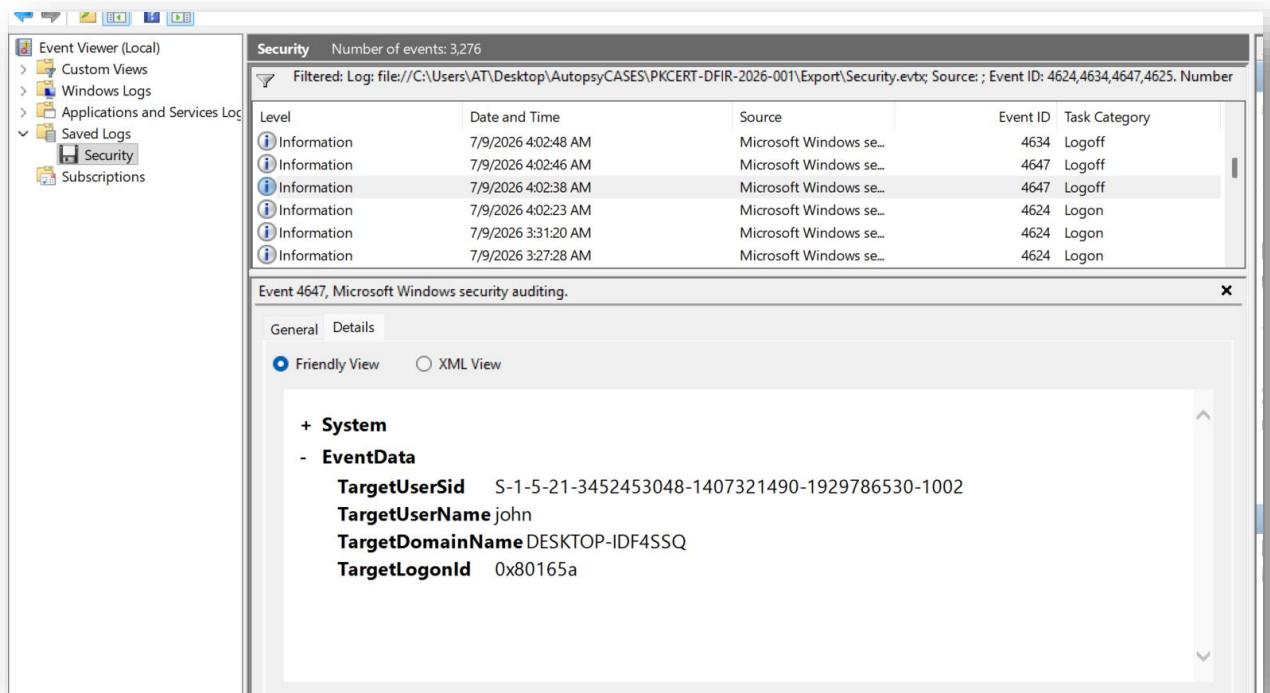
The logon event is recorded as **Event ID 4624** with **Logon Type 2 (Interactive Logon)**, indicating that the user logged into the computer locally using the Windows sign-in screen.

The logon session was assigned **Logon ID 0x80165a**.

Later, **Event ID 4647** was recorded at **09 July 2026 at 04:02 AM (Local Time)**. This event indicates that user **john** manually initiated a logoff from the system. The event contains the same **Logon ID (0x80165a)**, confirming that it belongs to the same user session.

A corresponding **Event ID 4634** for this session was not identified during the examination of the filtered Security log. However, the available evidence is sufficient to establish the beginning and the user-initiated end of this login session.





Event Summary

Date & Time (Local)	Event ID	Description	User	Logon ID
09 Jul 2026 – 03:16 AM	4624	Successful Interactive Logon	john	0x80165a
09 Jul 2026 – 04:02 AM	4647	User Initiated Logoff	john	0x80165a

Interpretation

The Security Event Log confirms that **john** started an interactive Windows session at **03:16 AM** and later initiated a manual logoff at **04:02 AM**. Both events share the same **Logon ID (0x80165a)**, demonstrating that they belong to the same login session. This session defines one period of user activity that will be correlated with other forensic artifacts such as Prefetch files, UserAssist, Recent Documents, browser history, and USB activity to reconstruct the user's actions during the investigation period.

5.3 Prefetch Analysis

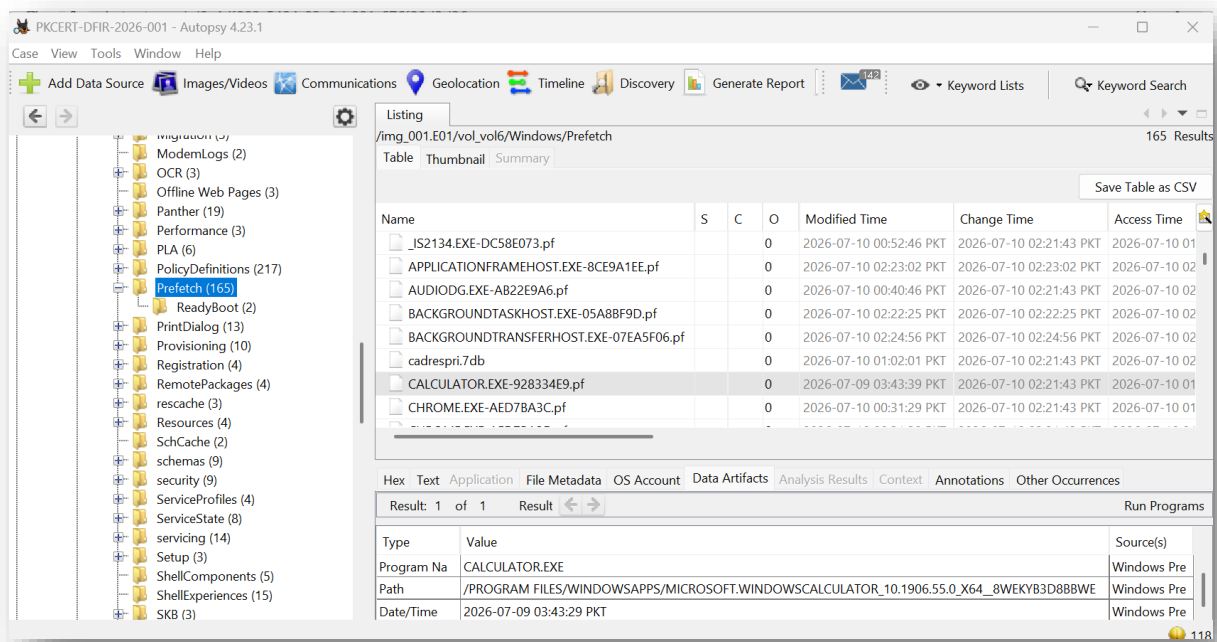
Purpose

Windows Prefetch records applications that have been executed on the system. During an investigation, Prefetch helps determine which programs the user ran and approximately when they were last executed. This artifact was examined to identify applications used by **John** during the suspected insider misuse incident.

Findings

The Prefetch directory contained execution records for both Windows system processes and user applications. The analysis focused only on applications related to user activity.

The following applications were identified as relevant:



Application	Modified Time (PKT)	Observation
WHOAMI.EXE	10 Jul 2026, 12:23 AM	User executed the whoami command from Command Prompt.
POWERSHELL.EXE	10 Jul 2026, 12:53 AM	PowerShell was launched by the user.
NOTEPAD.EXE	10 Jul 2026, 12:28 AM	Notepad was executed to open or edit text documents.
MSPAINT.EXE	09 Jul 2026, 03:44 AM	Microsoft Paint was executed.
TREE.COM	10 Jul 2026, 12:23 AM	The tree command was executed in Command Prompt.
NETSH.EXE	10 Jul 2026, 12:43 AM	Network-related commands were executed.
WPS.EXE	10 Jul 2026, 12:29 AM	WPS Office was opened to access office documents.
MSEDGE.EXE	10 Jul 2026, 02:23 AM	Microsoft Edge browser was executed.

Interpretation

The Prefetch records confirm that the suspect actively used several applications during the investigation period. The execution of **Command Prompt**, **PowerShell**, and utilities such as **WHOAMI** and **TREE** indicates command-line activity. The presence of **Notepad** and **WPS Office** suggests that text files and office documents were accessed. Browser execution (Microsoft Edge) indicates internet activity during the same period.

Although Prefetch confirms that these programs were executed, it does **not** identify which specific files were opened. Additional artifacts such as **RecentDocs**, **Jump Lists**, **LNK files**, and **UserAssist** are required to correlate application execution with individual files and user actions.

Evidence Summary

- User executed multiple desktop applications.
- Office documents were accessed using WPS Office.
- Command-line tools were used (WHOAMI, TREE, PowerShell).
- Web browser activity was present.
- The execution history is consistent with the timeline established from the Windows Security Event Logs.

Conclusion:

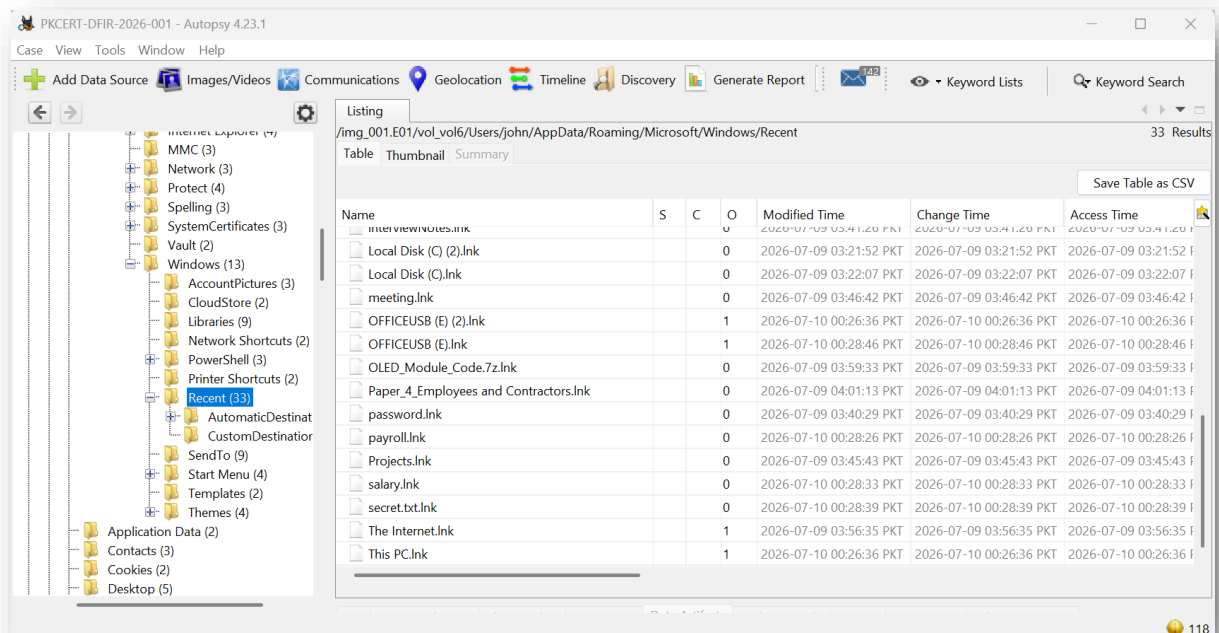
The Prefetch artifact confirms that John executed several applications relevant to the investigation, including WPS Office, Notepad, Microsoft Edge, Command Prompt utilities, and PowerShell. These findings support the reconstruction of user activity and provide evidence that the suspect actively interacted with documents, command-line tools, and web resources during the investigation period.

5.4 Recent Documents Analysis

Purpose

The Recent Documents artifact was examined to identify files and folders recently accessed by the user account **john** during the investigation period. Windows automatically creates shortcut (.lnk) files whenever users open documents or browse folders. These shortcuts help investigators determine user activity even if the original files have later been moved or deleted.

Finding



Analysis of the **Recent** folder located at:

`C:\Users\john\AppData\Roaming\Microsoft\Windows\Recent`

revealed numerous shortcut files created during the investigation period. The shortcuts indicate that the user accessed several finance-related documents, folders, and external storage locations.

The following items were considered most relevant to the investigation.

Date & Time (PKT)	Shortcut	Observation
09 Jul 2026 03:22 AM	company.lnk	Company folder accessed.
09 Jul 2026 03:22 AM	Finance.lnk	Finance directory opened.
09 Jul 2026 03:22 AM	HR.lnk	HR directory accessed.
09 Jul 2026 03:40 AM	password.lnk	Password file accessed.
09 Jul 2026 03:42 AM	Employelist.lnk	Employee list document opened.
09 Jul 2026 03:45 AM	finance_sketch.lnk	Finance-related document accessed.
09 Jul 2026 03:46 AM	meeting.lnk	Meeting document opened.
09 Jul 2026 03:54 AM	Downloads.lnk	Downloads folder accessed.
09 Jul 2026 04:01 AM	Paper_4_Employees and Contractors.lnk	PDF/document viewed.
10 Jul 2026 12:13 AM	Budget.lnk	Budget document opened.
10 Jul 2026 12:14 AM	expenses.lnk	Expense spreadsheet accessed.
10 Jul 2026 12:16 AM	salary.lnk	Salary document opened.
10 Jul 2026 12:20 AM	Finance.lnk	Finance folder revisited.
10 Jul 2026 12:26 AM	OFFICEUSB (E).lnk	External USB drive accessed.
10 Jul 2026 12:28 AM	payroll.lnk	Payroll document accessed.
10 Jul 2026 12:28 AM	confidential.txt.lnk	Confidential text file opened.
10 Jul 2026 12:28 AM	secret.txt.lnk	Secret document accessed.
10 Jul 2026 12:28 AM	Backup.lnk	Backup folder or file accessed.

Analysis

The Recent Documents artifact shows a clear progression of user activity.

During the first session on **9 July**, the user browsed company folders including **Company**, **Finance**, and **HR** before opening several documents related to employees, passwords, meetings, and finance.

A second group of shortcut files created on **10 July** shows the user returning to financial documents such as **Budget, Salary, Expenses, and Payroll**. The presence of **OFFICEUSB (E).lnk** indicates that an external USB storage device was accessed during the same period. Additional shortcuts referencing **confidential.txt, secret.txt, and Backup** suggest that sensitive files were viewed shortly before the USB activity.

These artifacts support the simulated incident scenario in which the suspect reviewed confidential company information and later copied selected files to removable media.

Conclusion

The Recent Documents artifact shows that user **john** interacted with multiple company folders and confidential documents during the investigation period. It also provides evidence that an external USB device was accessed after several financial and confidential documents were opened. These findings support the activity observed in other artifacts and strengthen the reconstruction of the user's actions.

5.5. Recycle Bin Analysis

Purpose

The purpose of this analysis was to determine whether the user attempted to delete any files during the incident period. Deleted files can indicate attempts to hide evidence or remove sensitive information before leaving the organization.

Findings

Autopsy identified one deleted file in the Recycle Bin.

Item	Value
Deleted File	Budget.xlsx
Original Location	C:\Company\Finance\Budget.xlsx
Deleted By	john
Deletion Time	10 July 2026, 12:30:36 AM PKT

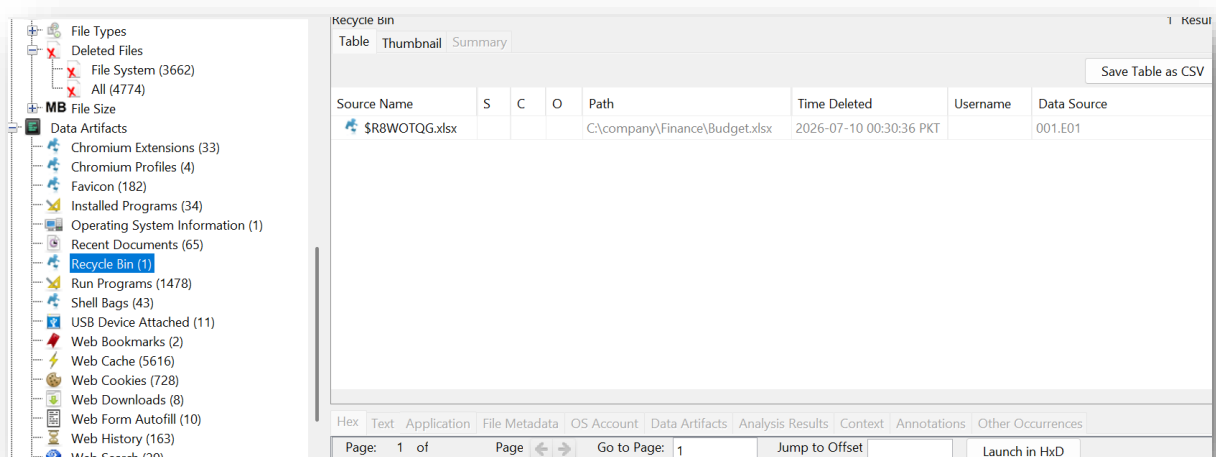
Analysis

The Recycle Bin artifact shows that the user **john** deleted the file **Budget.xlsx** from the Finance folder.

Although the file was deleted, its metadata remained available in the Recycle Bin, allowing investigators to determine its original location and the exact time of deletion.

Since this document was stored inside the Finance directory, it may have contained sensitive business information. The deletion occurred after the user had accessed multiple finance-related documents, suggesting an attempt to remove evidence before ending the session.

Evidence



5.6 USB Device Analysis

Purpose

The purpose of this analysis was to determine whether any removable storage device was connected to the system during the investigation period and whether it was associated with the user's activity involving company documents.

Findings

Analysis of the **SYSTEM** registry hive identified a registered USB storage device with the following information:

Item	Value
Vendor	PNY
Product	USB 3.2.1 Flash Drive
Registry Key	Disk&Ven_PNY&Prod_USB_3.2.1_FD&Rev_PMAP
Registry Hive	SYSTEM

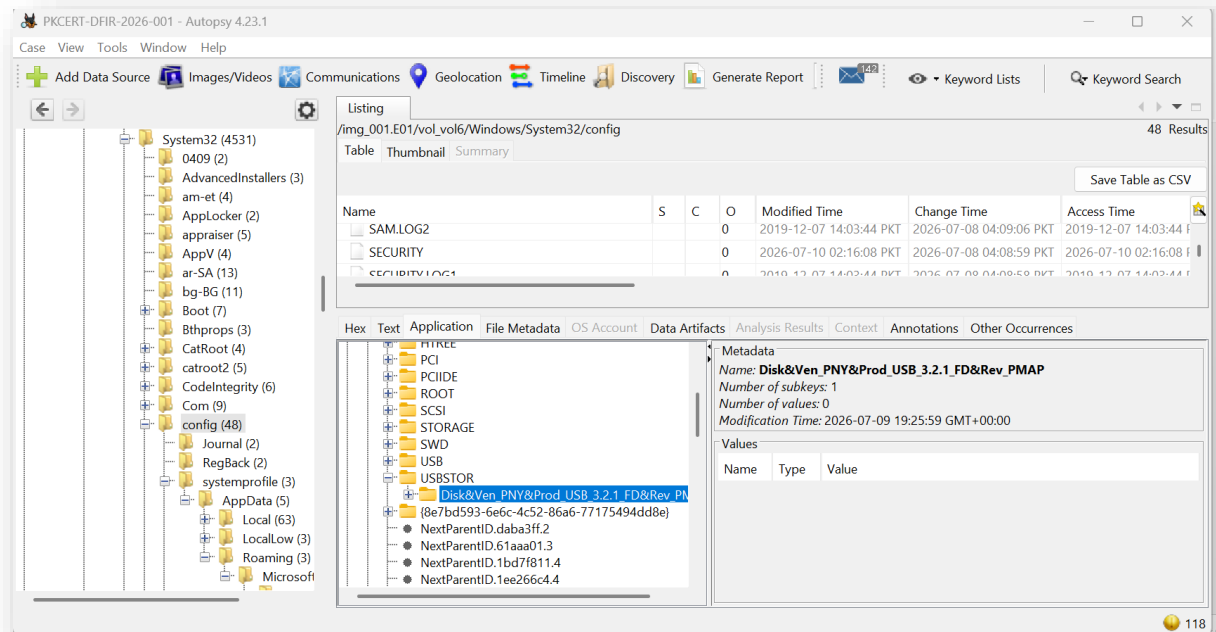
Further examination of the **Recent Documents** artifact revealed shortcut files named **OFFICEUSB (E).lnk** and **OFFICEUSB (E) (2).lnk**, indicating that the removable storage device mounted as drive **E:** was accessed by the user.

During the same period, shortcut files for several finance-related documents, including **Budget.xlsx**, **Salary.xlsx**, and **Payroll**, were also created or updated. The close timing between access to these documents and the removable storage device indicates that both were used during the same user session.

Although no artifact directly confirms that files were copied to the USB device, the combined registry and shortcut evidence is **consistent with the use of an external USB storage device while accessing sensitive company documents**.

Evidence Summary

Artifact	Observation
SYSTEM Registry	PNY USB 3.2.1 Flash Drive registered on the system
Recent Documents	OFFICEUSB (E).lnk and OFFICEUSB (E) (2).lnk found
Recent Documents	Budget.lnk, Salary.lnk, Payroll.lnk accessed during the same session
Correlation	USB access occurred during activity involving finance-related documents



5.7 Timeline of Events

Purpose

The timeline below was created by correlating evidence collected from Windows Security Event Logs, Recent Documents, Prefetch files, USB Registry artifacts, and the Recycle Bin. The purpose of this timeline is to reconstruct the user's activities during the investigation period.

Date & Time (PKT)	Artifact	Event
09 Jul 2026 03:16 AM	Security Log (Event ID 4624)	User john successfully logged on to the system using an interactive logon (Logon Type 2).
09 Jul 2026 03:22 AM	Recent Documents	User browsed the Company directory and accessed Finance and HR folders.
09 Jul 2026 03:40 AM	Recent Documents	Password-related document (password.lnk) was opened.
09 Jul 2026 03:42 AM	Recent Documents	Employee list document was accessed.
09 Jul 2026 03:45 AM	Recent Documents	Project folders and finance-related files were accessed.
09 Jul 2026 03:46 AM	Recent Documents	Meeting-related document was opened.
09 Jul 2026 03:54 AM	Recent Documents	Downloads folder and PDF document were accessed.
09 Jul 2026 03:56 AM	Recent Documents	Internet shortcut indicates web browsing activity.
09 Jul 2026 03:59 AM	Recent Documents	ZIP archive (OLED_Module_Code.7z) was accessed.
09 Jul 2026 04:01 AM	Recent Documents	Additional PDF document was opened.
09 Jul 2026 04:02 AM	Security Log (Event ID 4647)	User john initiated logoff, ending the interactive session.
09 Jul 2026 07:25 PM (UTC) / 10 Jul 2026 12:25 AM (PKT)	USB Registry	PNY USB storage device registry key was updated, indicating USB storage connection.
10 Jul 2026 12:13 AM	Recent Documents	Budget document accessed again.
10 Jul 2026 12:14 AM	Recent Documents	Expenses document accessed.
10 Jul 2026 12:16 AM	Recent Documents	Salary document accessed.
10 Jul 2026 12:18 AM	Recent Documents	HR folder revisited.
10 Jul 2026 12:20 AM	Recent Documents	Finance folder revisited.
10 Jul 2026 12:26 AM	Recent Documents	USB drive (OFFICEUSB (E:)) accessed.
10 Jul 2026 12:28 AM	Recent Documents	Salary, Payroll and Secret documents were accessed from recent items.
10 Jul 2026 12:30 AM	Recycle Bin	Budget.xlsx deleted and moved to the Recycle Bin.
10 Jul 2026 12:40 AM	Recent Documents	FTK Imager executed during evidence acquisition.

Timeline Analysis

The reconstructed timeline shows that the user **john** logged into the system on **9 July 2026** and accessed multiple finance-related documents, folders, downloaded files, and company data before logging off. During the second period of activity, evidence shows that a USB storage device manufactured by **PNY** was connected, after which finance-related documents were accessed again. Shortly afterward, **Budget.xlsx** was deleted and placed into the Recycle Bin. These events are consistent with the simulated incident scenario in which the user accessed confidential company information, interacted with removable storage, and attempted to remove evidence before the forensic image was acquired.

6. Findings

The examination identified that user **john** was the primary user during the investigation period. Security logs confirmed a successful interactive logon followed by a user-initiated logoff. Prefetch analysis showed execution of several applications including WPS Office, PowerShell, Command Prompt, Paint, and Windows utilities. Recent Documents artifacts indicated access to multiple finance-related documents and folders. Evidence also showed that a removable USB device named **OFFICEUSB (E:)** was accessed. Finally, Recycle Bin artifacts confirmed that **Budget.xlsx** was deleted from the Finance directory during the investigation period.

7. Conclusion

The forensic examination successfully reconstructed the activities performed by user **john** during the selected time period. The collected evidence demonstrates that the user accessed company finance documents, executed multiple system and productivity applications, interacted with a removable USB storage device, and deleted at least one finance-related document. By correlating Event Logs, Prefetch files, Recent Documents, and Recycle Bin artifacts, the investigation established a consistent timeline of user activity. No conclusions are made regarding intent; the report presents only the observable digital evidence recovered from the forensic image.